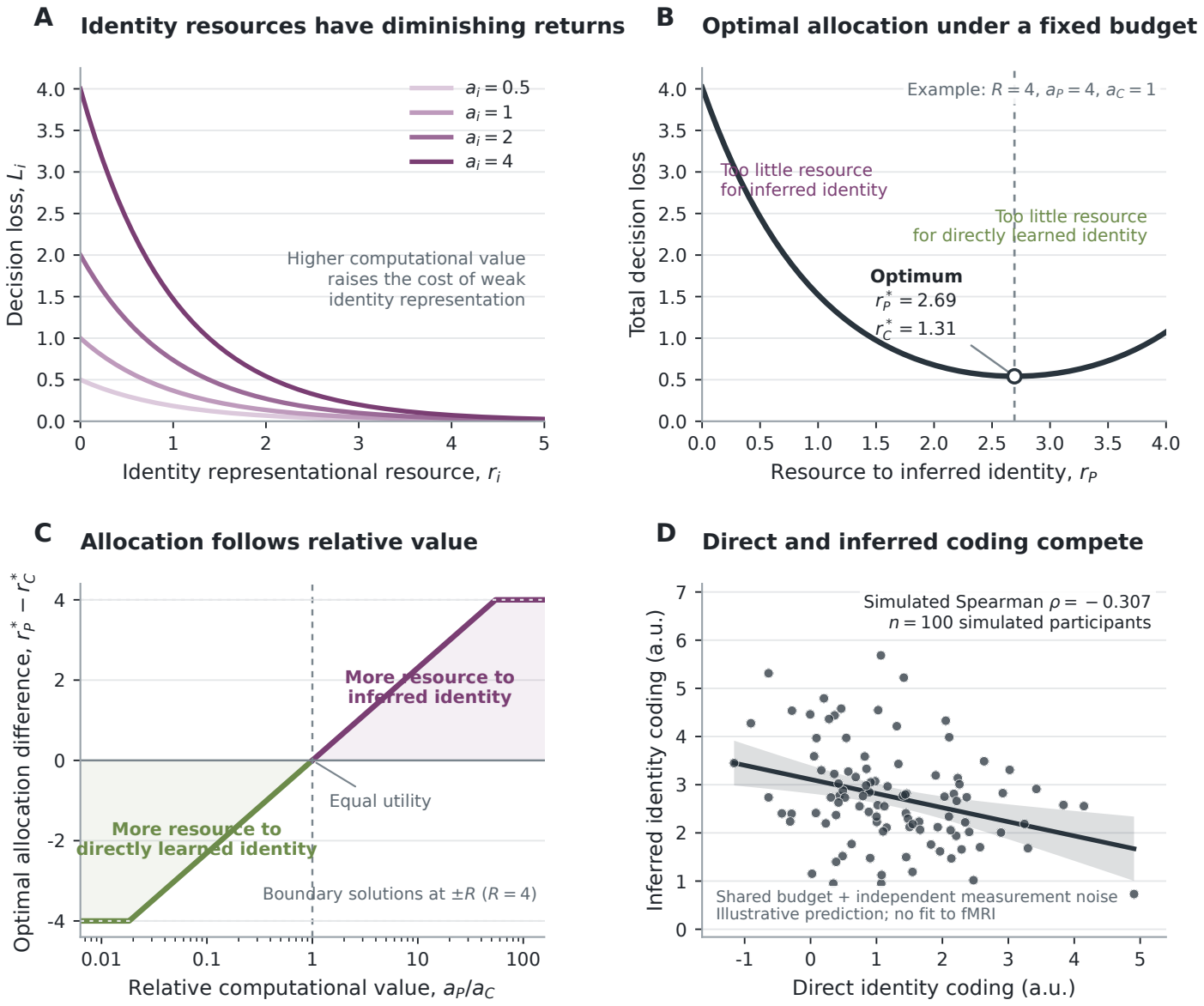
A
Identity resources have diminishing returns
$a_i = 0.5$
$a_i = 1$
$a_i = 2$
$a_i = 4$
Higher computational value raises the cost of weak identity representation
Decision loss, L_i
Identity representational resource, r_i
B
Optimal allocation under a fixed budget
Example: $R = 4$, $a_P = 4$, $a_C = 1$
Too little resource for inferred identity
Too little resource for directly learned identity
Optimum
$r_P^* = 2.69$
$r_C^* = 1.31$
Total decision loss
Resource to inferred identity, r_P
C
Allocation follows relative value
More resource to inferred identity
Equal utility
More resource to directly learned identity
Boundary solutions at $\pm R$ ($R = 4$)
Optimal allocation difference, $r_P^* - r_C^*$
Relative computational value, a_P/a_C
D
Direct and inferred coding compete
Simulated Spearman $\rho = -0.307$
$n = 100$ simulated participants
Shared budget + independent measurement noise
Illustrative prediction; no fit to fMRI
Inferred identity coding (a.u.)
Direct identity coding (a.u.)